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Structured Notes:
Skim Milk Masquerades as Cream

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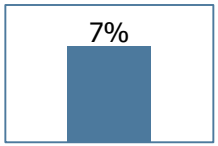
Structured Notes

- Structured notes promise intriguing outcomes:
  - Partial downside protection with upside participation

- or –

  - Above-market yield



- But what do they really deliver?
- And with what costs and risks?

The Pitch: Equity Upside with Limited Downside

A Recent Five-Year Structured Note Tied to the S&P 500 Price Return

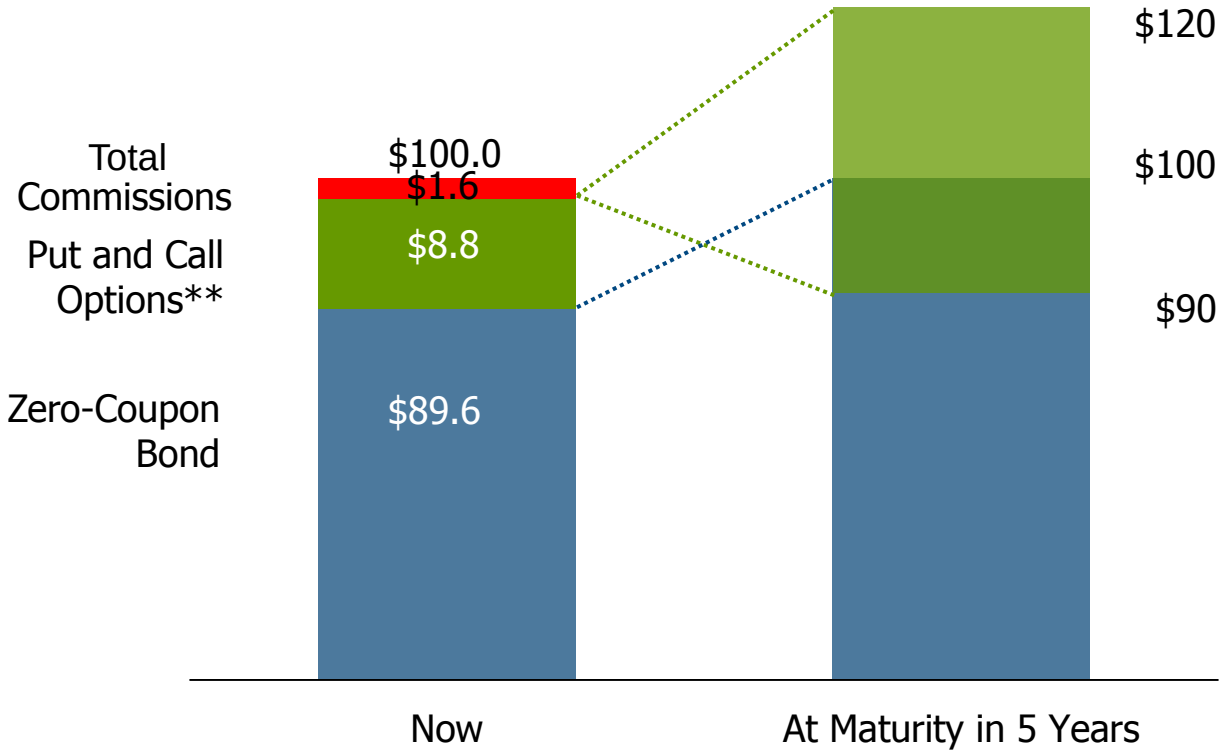
The Pitch		The Fine Print
Upside	S&P 500 price return	No yield
Partial protection	No loss on first 28% cumulative drop*	No dividend
		Need to wait five years for payoff

*Assuming issuer makes payment

Data do not represent past performance and are not a promise of actual future results or a range of future results.

How It Works: Downside-Protected Structured Note

Bank Packages Zero-Coupon Bond with Options on S&P 500



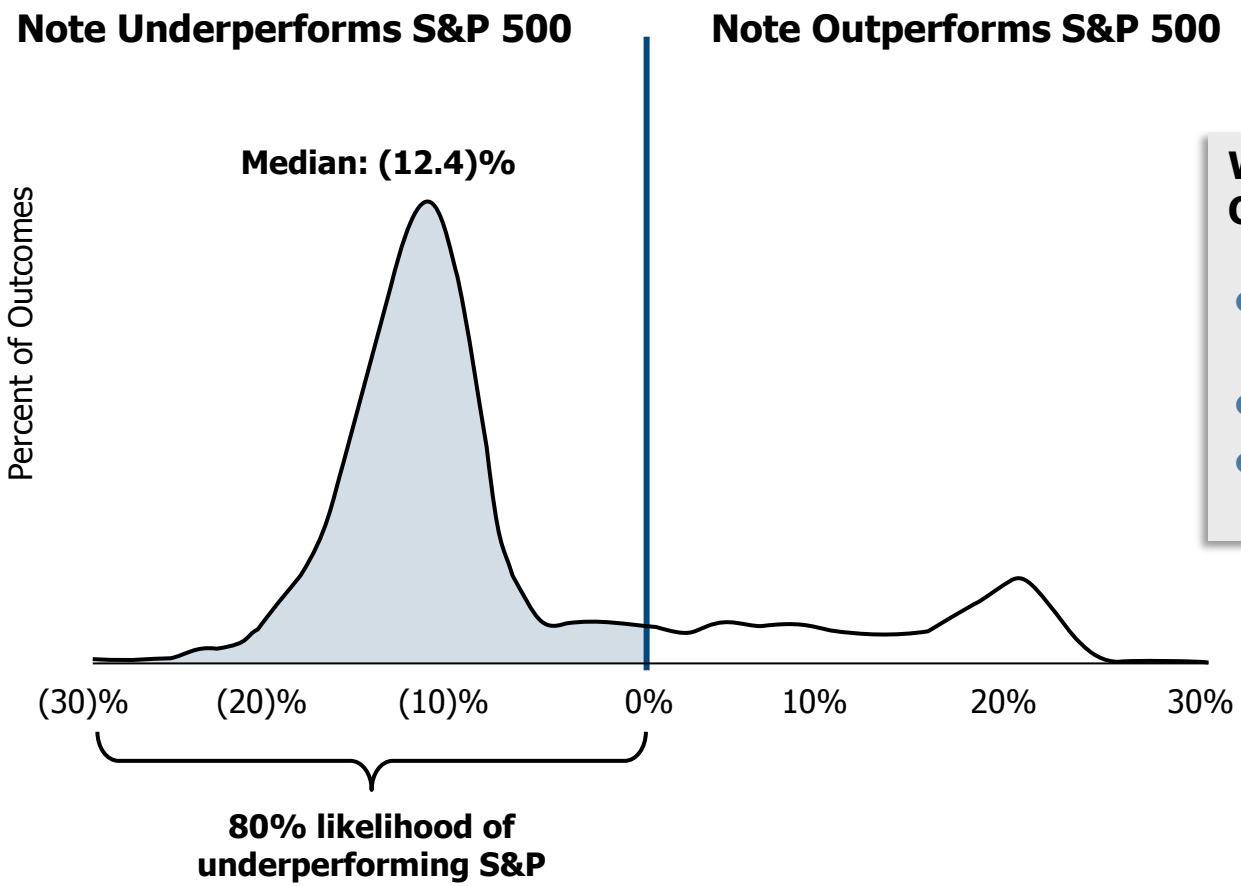
Illustrative Outcomes

Payoff*	**....if S&P Price**
$120	+20%
$100	0% to (28)%
$90	(38)%

*Assuming issuer makes payment. Depending on S&P level, returns could be higher or lower than shown.
**Assuming no spread on options
Hypothetical estimate for pretax total returns at maturity based on market pricing on March 18.,2015. **Data do not represent past performance and are not a promise of actual future results or a range of future results.**
Source: Standard & Poor's and Bernstein

A Good Equity Replacement? Clearly Not!

Projected Distribution of Returns Relative to S&P 500*



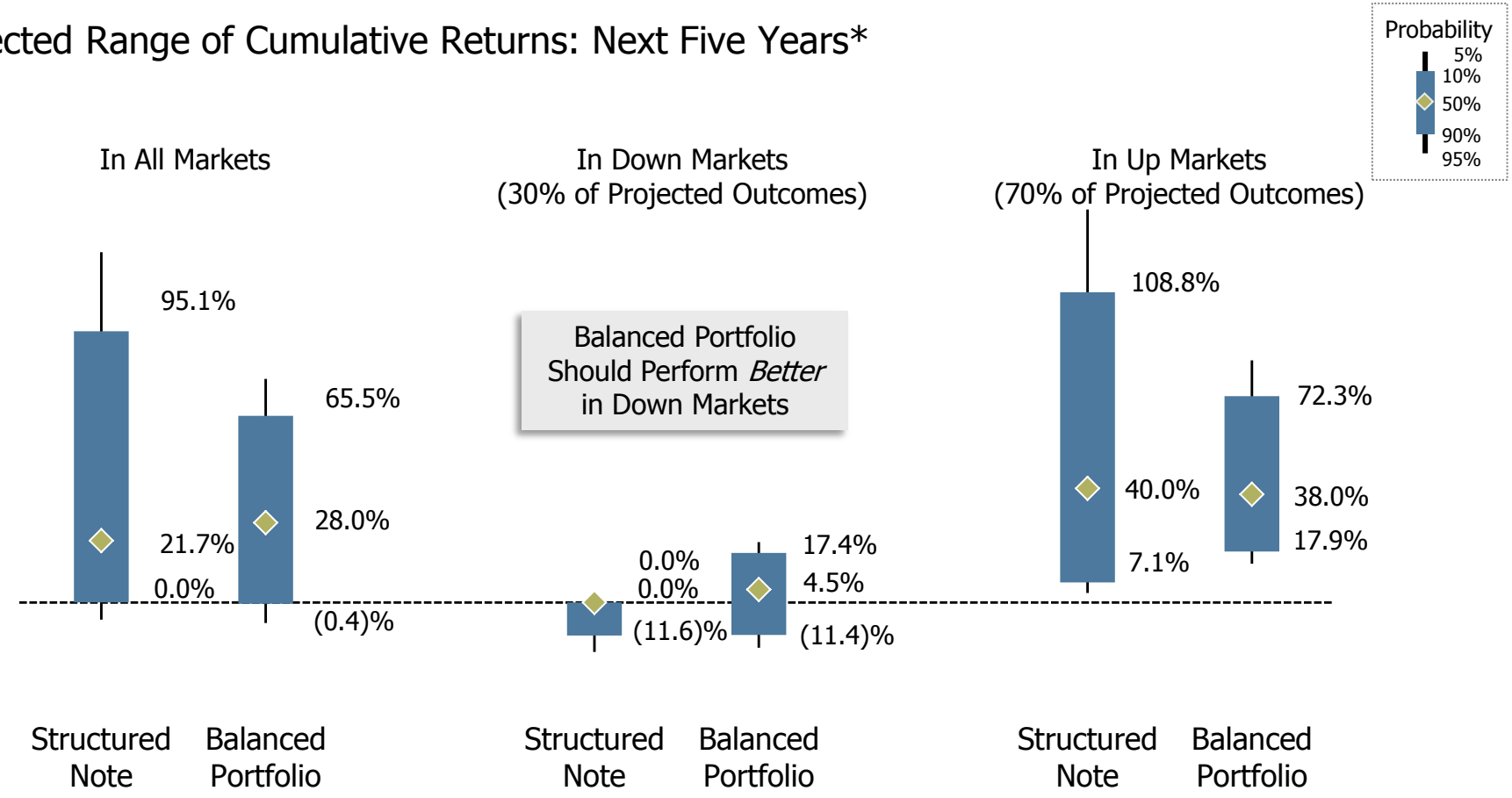
Why Expected Outcomes Are Poor:

- No dividends for five years
- Commissions
- Bond and option spreads

*Measured as cumulative pretax return for structured note (as outlined and assuming the issuer makes payment) relative to a direct investment in the S&P 500 Index over a five-year period. Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next five years. **Data do not represent past performance and are not a promise of actual future results or a range of future results**. See Notes on Wealth Forecasting System in the Appendix.
Source: Bernstein

A Good Balanced Portfolio Alternative? Clearly Not!

Projected Range of Cumulative Returns: Next Five Years*

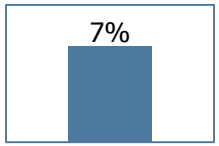


*Measured as cumulative pretax returns for direct investments and structured note (as outlined and assuming the issuer makes payment) over a five-year period. Up/down markets reflect price appreciation/depreciation for S&P 500 Index over the period. Balanced portfolio modeled as 50.1% global equities, 2.7% real assets, 12.2% hedge funds, 32.5% intermediate-term diversified municipals, and 2.5% inflation-sensitive municipals. Global equities modeled as 21% US diversified, 21% US value, 21% US growth, 7% US small-/mid-cap, 22.5% developed international and 7.5% emerging-market stocks

Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next five years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix .

Source: Bernstein

The Pitch: Where Else Can You Get a 7% Yield?



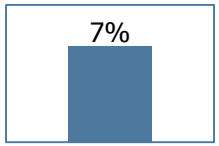
A Recent 10-Year Note with a 7% Yield

The Pitch	
Annual coupon	7%
Maturity	10 Years

The Fine Print
If both the Russell 2000 and EAFE Indexes are above 70% of initial levels
Automatically called (starting in Year 2) if both indexes are above starting levels on any coupon date
Receive par at maturity, unless one index has fallen by 50% or more; if so, the note incurs same loss

Data do not represent past performance and are not a promise of actual future results or a range of future results.
Source: Notes offer as of February 2015

How It Works: The Above-Market-Yield Note

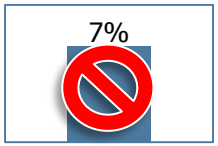


- Bank packages a bond and a series of complex options
- Each year, the investor will get a payout as follows:

	Scenario*	A Both indexes above starting level	B Neither index declines more than 30%	C One or both indexes decline more than 30% but neither more than 50%	D One or both indexes decline more than 50%
Outcomes	**Annual Coupon**	7%	7%	0%	0%
	Note Called?	Yes, beginning in Year 2	No	No	No
	Principal Payment at Call or Maturity	Par	Par	Par	Lowest index price return

*Index levels measured at scheduled quarterly (1.75%) coupon payment dates. Note can be called only on a coupon date. Assumes issuer makes payments. In our analysis, we have modeled annual coupon payments and call dates. **Data do not represent past performance and are not a promise of actual future results or a range of future results**.
Source: Notes offering and Bernstein

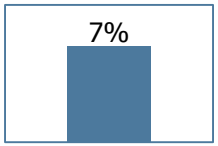
If Yield Is 7%, It Won't Be a 10-Year Note; If It's a 10-Year Note, It Won't Yield 7%



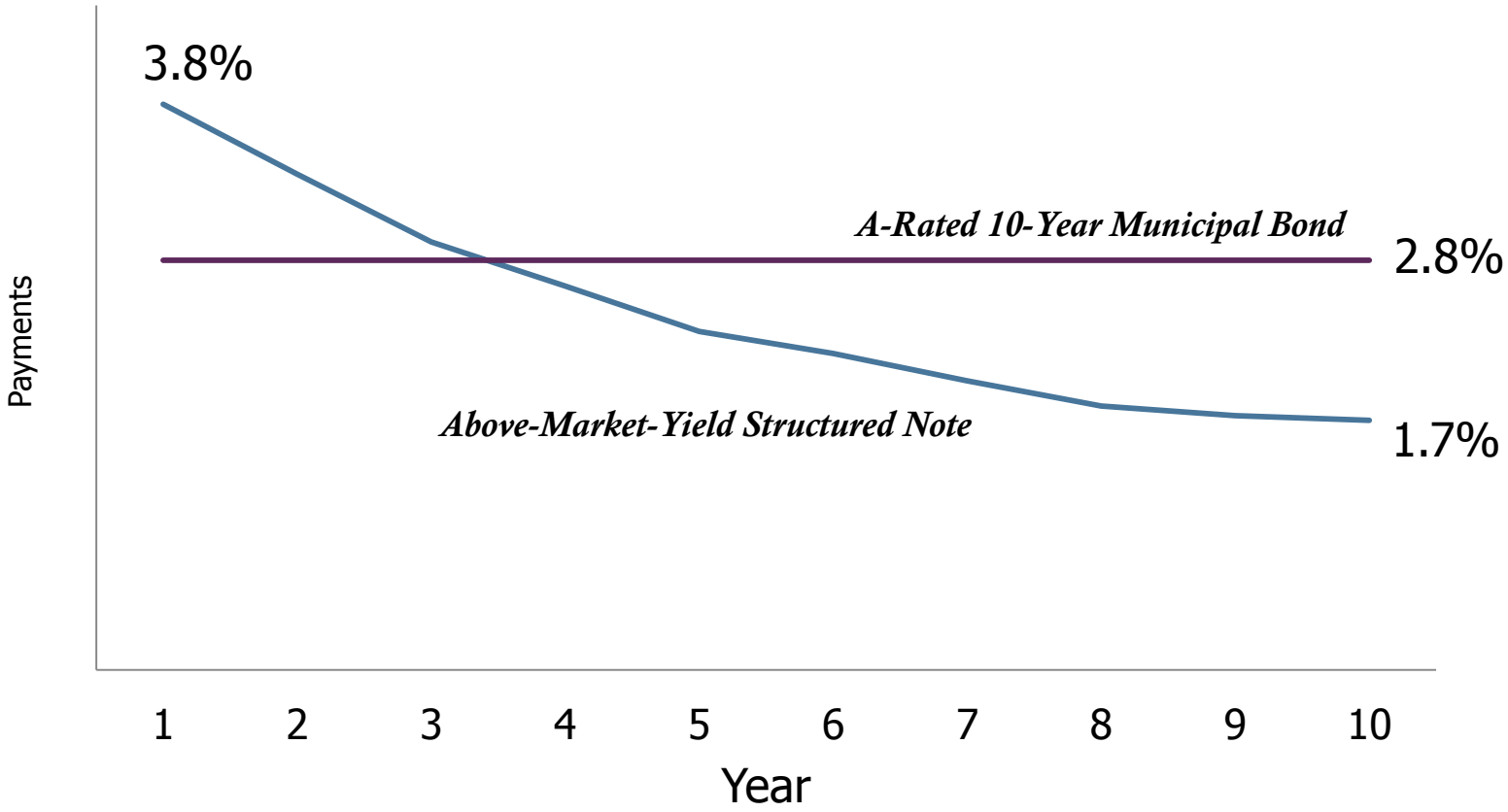
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Probability Note Called (Cumulative)	47%	62%	70%	75%	79%	82%	84%	85%	86%	100%
Probability Note Pays 0% Yield (in this year)	3%	15%	27%	34%	42%	46%	51%	55%	57%	57%
Expected Average Payments (pretax)	6.8%	5.9%	5.1%	4.6%	4.0%	3.8%	3.5%	3.2%	3.0%	3.0%
Expected Average Payments (after-tax)	3.8%	3.4%	2.9%	2.6%	2.3%	2.1%	2.0%	1.8%	1.7%	1.7%

Assumes issuer makes payments. Our analysis modeled annual coupon payments and call dates. Probability Note Pays 0% Yield applies to scenarios when the note has not been called. Expected Average Yield (pretax) calculated as Probability Note Pays 7% Yield times 7%. Expected Average Yield (after-tax) calculated as Expected Average Yield (pretax) less ordinary income taxes, assuming top marginal federal tax rates of 43.4%.
Based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix.
Source: Bernstein

Better than Municipal Bonds? Not Really

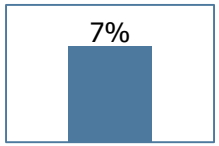


Expected Average After-Tax Payments



Assumes issuers make payments. In our analysis, we have modeled annual coupon payments and call dates. Above-Market-Yield Structured Note calculated as Expected Average Yield (pretax) less ordinary income taxes, assuming top marginal federal tax rates (43.4%). A-Rated 10-Year Municipal Bond based on Muni 10-Year Single-A General Obligation Yield as of March 31, 2015.
Based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results**. See Notes on Wealth Forecasting System in the Appendix.
Source: Bloomberg and Bernstein

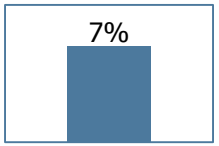
A Good Muni Replacement? Clearly Not!



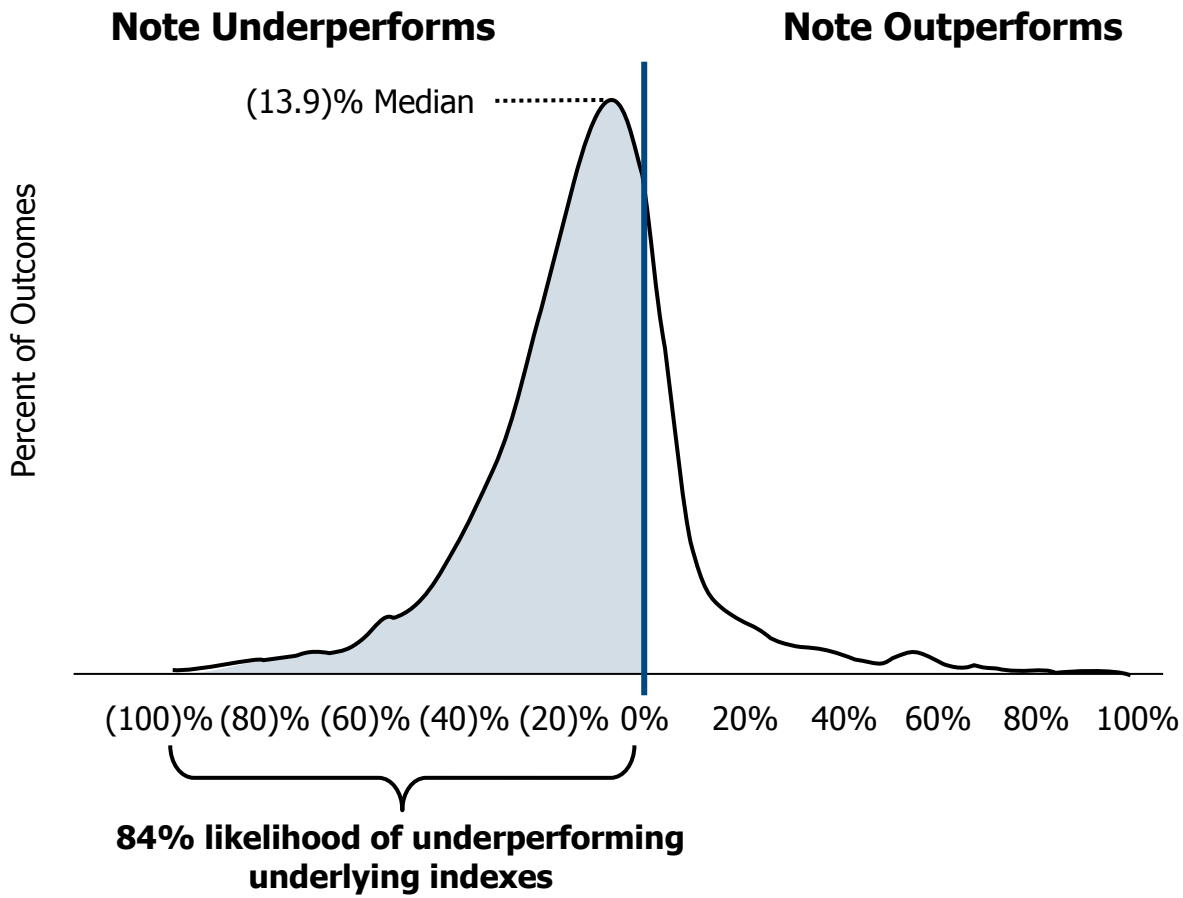
	Above-Market-Yield Note	Municipal Bond
Variability of Coupon	High	None
Tax on Yield	High	None
Illiquidity	High	Low
Principal Risk	Moderate	Low
Term Uncertainty	High	Low
Correlation to Equities	High	Low or Negative

Data do not represent past performance and are not a promise of actual future results or a range of future results.
Source: Bernstein

A Good Equity Replacement? Clearly Not!



Projected Distribution of Cumulative Returns vs. Underlying Indexes*



Why Expected Outcomes Are Poor:

- Tied to worst of two volatile indexes
- Commissions
- Option and bond spreads
- If stock markets do well, note will be called
- If stock markets do poorly, note will continue and yield is increasingly likely to be 0.

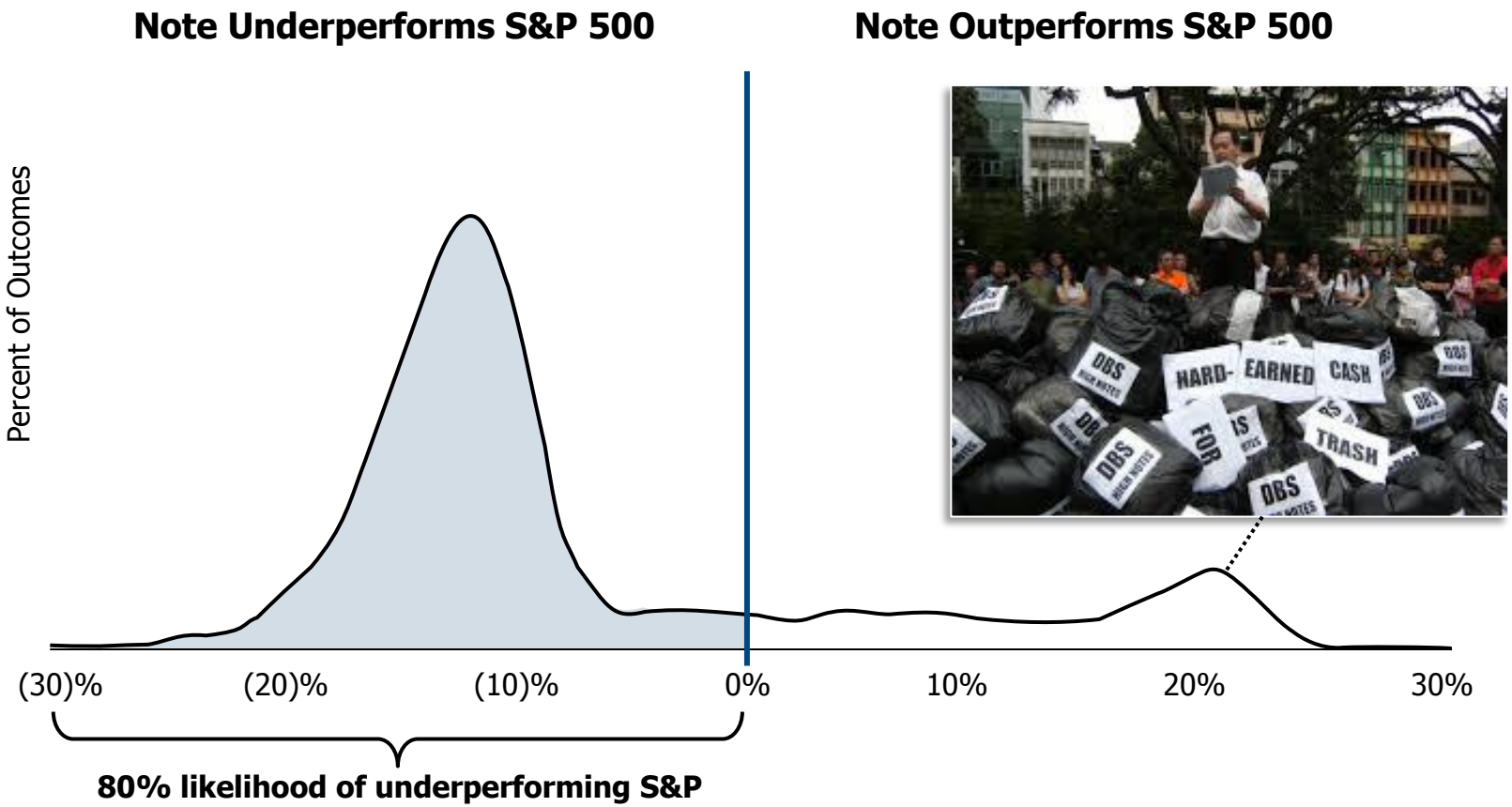
*Measured as cumulative pretax return for the structured note (as outlined and assuming the issuer makes payments) relative to a direct investment strategy allocated 50% to the Russell 2000 Index and 50% to the MSCI EAFE Index (rebalanced annually) over the investment term (based on implied term for an investment in the structured note over the period)
Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results**. See Notes on Wealth Forecasting System in the Appendix.
Source: Bernstein

The Problems with Structured Notes

- Expensive for a passive investment
  - Favorable funding for issuer, selling commissions, option markups
  - Forgo dividend income
- Potentially risky
  - Illiquidity: Sales before maturity likely to be at unfavorable prices
  - May lose substantial principal, depending on note terms
  - Buyer bears issuer's credit risk
- Optically attractive but inefficient
  - Yield, if any, taxed as ordinary income
  - Implied interest (not paid) may be taxed as ordinary income if note provides full downside protection
  - Our research shows investors typically better-off owning underlying bonds or equities and managing risk

Heads They Win, Tails You Lose?
Issuer Default Is More Likely When Protection Is Needed

Projected Distribution of Returns Relative to S&P 500*



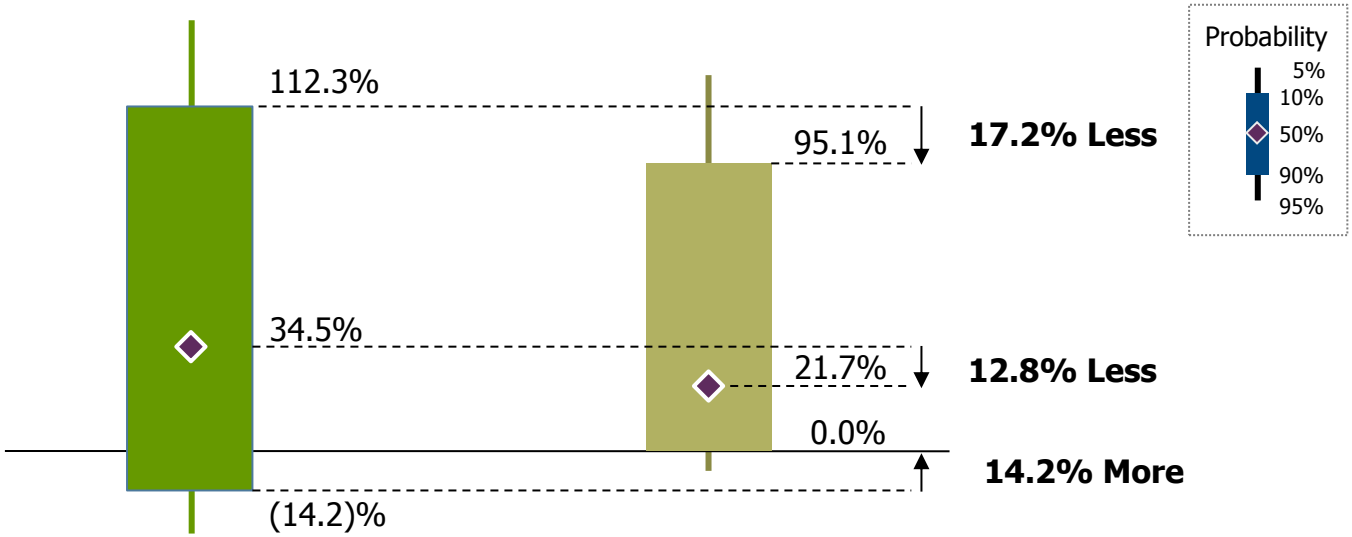
*Measured as cumulative pretax return for structured note (as outlined and assuming the issuer makes payment) relative to a direct investment in the S&P 500 Index over a five-year period. Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next five years. **Data do not represent past performance and are not a promise of actual future results or a range of future results**. See Notes on Wealth Forecasting System in the Appendix .

Source: Bernstein

Appendix

Paying a Lot for Little Protection

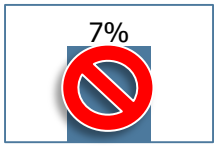
Projected Range of Cumulative Returns: Next Five Years*

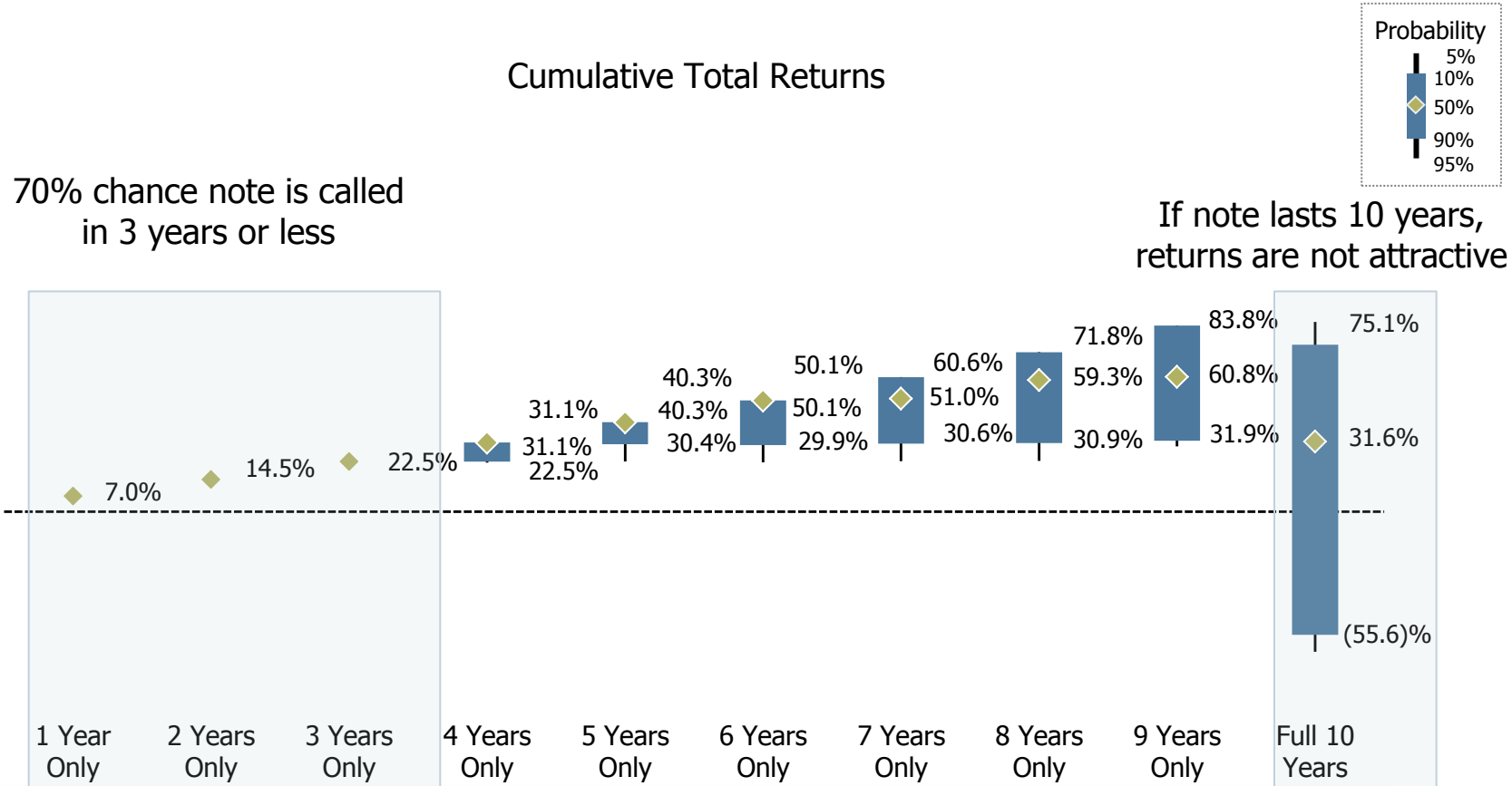


	S&P 500 Total Return	Downside Protection Note
Probability of a Loss	**20%**	**8%**

*Measured as cumulative pretax returns for direct investment in the S&P 500 Index and structured note (as outlined and assuming the issuer makes payment) over a five-year period. Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next five years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix.
Source: Bernstein

If Yield Is 7%, It Won't Be a 10-Year Note; If It's a 10-Year Note, It Won't Yield 7%



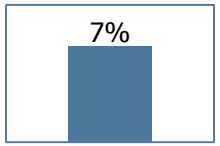


	1 Year Only	2 Years Only	3 Years Only	4 Years Only	5 Years Only	6 Years Only	7 Years Only	8 Years Only	9 Years Only	Full 10 Years
% of Outcomes	**47%**	**15%**	**8%**	5%	4%	3%	2%	1%	1%	**14%**
Cumulative	**47%**	**62%**	**70%**	75%	79%	82%	84%	85%	86%	**100%**

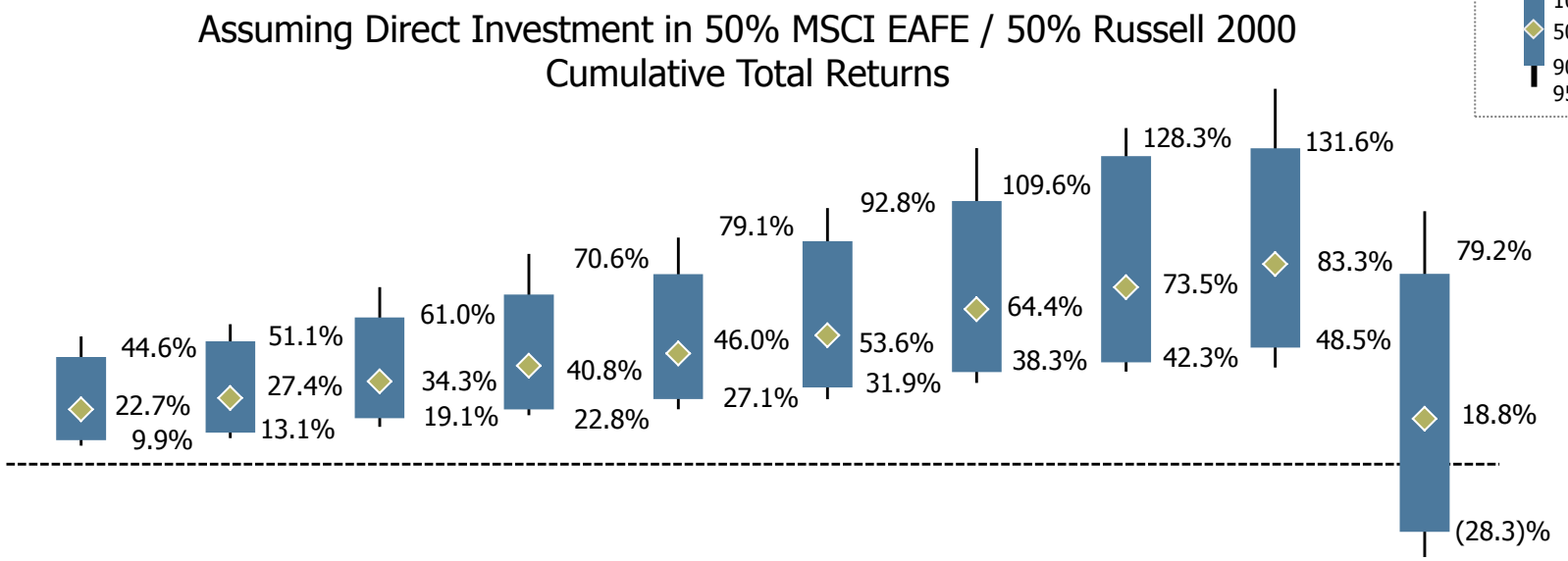
Measured as cumulative pretax return for structured note (as outlined and assuming the issuer makes payments) over the term of the note. For comparison purposes, we assume annual coupon payments are reinvested at the IRR (Internal Rate of Return) of the note. Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix.

Source: Bernstein

What if You Owned the Underlying Indexes Instead?



Projected Outcomes by Year from Owning the Underlying Indexes



	1 Year Only	2 Years Only	3 Years Only	4 Years Only	5 Years Only	6 Years Only	7 Years Only	8 Years Only	9 Years Only	Full 10 Years
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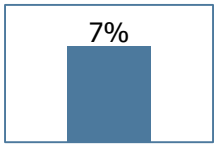
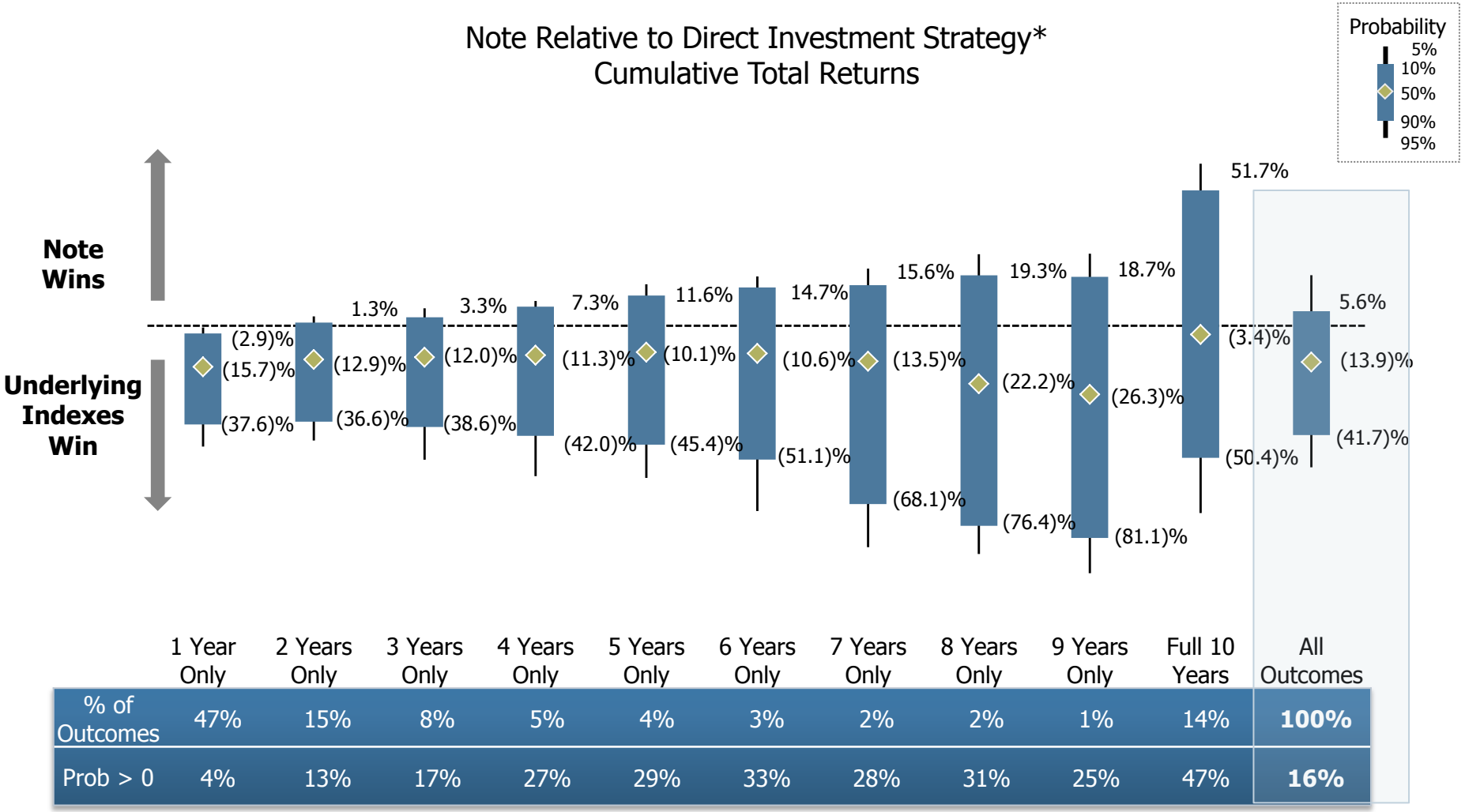
Cumulative	**47%**	**62%**	**70%**	75%	79%	82%	84%	85%	86%	**100%**

Measured as cumulative pretax return for a direct investment allocated 50% to the Russell 2000 Index and 50% to the MSCI EAFE Index (rebalanced annually). Investment term calculated based on implied term for an investment in the structured note over the period. Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix.
Source: Bernstein

Above-Market Yield Note Lags Underlying Indexes



Note Relative to Direct Investment Strategy*
Cumulative Total Returns



	1 Year Only	2 Years Only	3 Years Only	4 Years Only	5 Years Only	6 Years Only	7 Years Only	8 Years Only	9 Years Only	Full 10 Years	All Outcomes
% of Outcomes	47%	15%	8%	5%	4%	3%	2%	2%	1%	14%	**100%**
Prob > 0	4%	13%	17%	27%	29%	33%	28%	31%	25%	47%	**16%**

*Measured as cumulative pretax return for the structured note (as outlined and assuming the issuer makes payments) relative to a direct investment strategy over the investment term (based on implied term for an investment in the structured note over the period). Pretax returns based on Bernstein Wealth Forecasting System estimates of the range of returns for the applicable capital markets over the next 10 years. **Data do not represent past performance and are not a promise of actual future results or a range of future results.** See Notes on Wealth Forecasting System in the Appendix.

Source: Bernstein

A Good Muni Replacement? Clearly Not!

	Above-Market-Yield Note	Municipal Bond
Variability of Coupon	High: 0% to 7%	None
Tax on Yield	High: Ordinary income	None: Tax-exempt
Illiquidity	High: Limited market for sales prior to call/maturity	Low but subject to market fluctuations
Principal Risk	Moderate: Issuer credit risk; downside risk in extremely poor equity market environments	Low: Issuer credit risk
Term Uncertainty	High: 1 to 10 years; likely 1–3 *or* 10 years	Low: 10-year fixed term
Correlation to Equities	High: In poor equity markets potentially no cash flow or access to principal for spending/rebalancing; may lead to selling out of (or failing to rebalance into) equities at worst possible time	Low (or Negative): Cash flow from coupon payments and access to principal (with potential for appreciation in poor equity markets)

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Source: Bernstein

Capital-Market Projections

	Median 10-Year Growth Rate	Mean Annual Return	Mean Annual Income	One-Year Volatility	10-Year Annual Equivalent Volatility
Cash Equivalents	1.6%	1.9%	1.9%	0.3%	3.6%
Short-Term Treasuries	2.3%	2.5%	2.4%	0.7%	3.4%
Short-Term Taxables	2.8%	2.9%	3.2%	1.0%	3.4%
Short-Term Diversified Municipals	1.6%	1.7%	1.7%	0.6%	2.4%
Int.-Term Treasuries	2.2%	2.3%	3.0%	4.2%	3.0%
Int.-Term Taxables	2.5%	2.6%	4.0%	4.6%	3.4%
Int.-Term Corporates	2.6%	2.8%	4.6%	5.5%	4.0%
Int.-Term Diversified Municipals	2.3%	2.4%	2.4%	3.6%	2.9%
Global Int.-Term Taxables (Hedged)	1.8%	2.0%	2.7%	4.0%	3.6%
Int.-Term TIPS	2.4%	2.8%	3.0%	2.9%	6.4%
High Yield	3.8%	4.5%	8.8%	12.5%	8.2%
Global Large-Cap (Unhedged)	6.7%	8.1%	2.6%	15.8%	14.7%
US Diversified	6.0%	7.6%	2.2%	16.4%	15.2%
US Value	6.4%	7.9%	2.7%	16.0%	15.0%
US Growth	5.7%	7.5%	1.8%	18.2%	16.7%
US Mid-Cap	6.4%	8.2%	1.9%	17.9%	16.9%
US Small-/Mid-Cap	6.4%	8.3%	1.9%	18.7%	17.7%
US Small-Cap	6.3%	8.6%	1.8%	20.5%	19.5%
Developed International	7.1%	9.0%	3.1%	18.0%	16.8%
Emerging Markets	5.3%	9.0%	3.1%	26.1%	25.6%
Global REITs	5.8%	7.4%	3.8%	16.9%	15.7%
Real Assets	5.5%	6.5%	2.9%	13.1%	14.0%
Diversified Hedge Fund	4.8%	5.1%	2.0%	11.0%	14.9%
Inflation	2.0%	2.3%	n/a	1.1%	5.4%

Based on 10,000 simulated trials, each consisting of 10-year periods. Reflects Bernstein's estimates and the capital-market conditions as of December 31, 2014. For hedge-fund asset classes, "Mean Annual Income" represents income and short-term capital gains.

Data do not represent past performance and are not a promise or a range of future results.

Source: Bernstein

Notes on Wealth Forecasting System

1. **Purpose and Description of Wealth Forecasting Analysis**

   Bernstein's Wealth Forecasting Analysis is designed to assist investors in making their long-term investment decisions as to their allocation of investments among categories of financial assets. Our planning tool consists of a four-step process: (1) Client-Profile Input: the client's asset allocation, income, expenses, cash withdrawals, tax rate, risk-tolerance level, goals, and other factors; (2) Client Scenarios: in effect, questions the client would like our guidance on, which may touch on issues such as when to retire, what his/her cash-flow stream is likely to be, whether his/her portfolio can beat inflation long-term, and how different asset allocations might affect his/her long-term security; (3) The Capital-Markets Engine: our proprietary model that uses our research and historical data to create a vast range of market returns, which takes into account the linkages within and among the capital markets, as well as their unpredictability; and (4) A Probability Distribution of Outcomes: based on the assets invested pursuant to the stated asset allocation, 90% of the estimated ranges of returns and asset values that the client could expect to experience are represented within the range established by the 5th and 95th percentiles on "box-and-whiskers" graphs. However, outcomes outside this range are expected to occur 10% of the time; thus, the range does not establish the boundaries for all outcomes. Expected market returns on bonds are derived taking into account yield and other criteria. An important assumption is that stocks will, over time, outperform long bonds by a reasonable amount, although this is in no way a certainty. Moreover, actual future results may not meet Bernstein's estimates of the range of market returns, as these results are subject to a variety of economic, market, and other variables. Accordingly, the analysis should not be construed as a promise of actual future results, the actual range of future results, or the actual probability that these results will be realized. The information provided here is not intended for public use or distribution beyond our private meeting.

Notes on Wealth Forecasting System

2. Modeled Asset Classes

The following assets or indexes were used in this analysis to represent the various model classes:

Asset Class	Modeled As	Annual Turnover
Cash Equivalents	3-month US Treasury bills	100%
Short-Term Treasuries	US Treasuries of 2-year maturity	50%
Short-Term Taxables	Taxable bonds of 2-year maturity	50%
Short-Term Diversified Municipals	AA-rated diversified municipal bonds of 2-year maturity	50%
Int.-Term Treasuries	US Treasuries of 7-year maturity	30%
Int.-Term Taxables	Taxable bonds of 7-year maturity	30%
Int.-Term Corporates	US investment-grade corporate debt of 7-year maturity	30%
Int.-Term Diversified Municipals	AA-rated diversified municipal bonds of 7-year maturity	30%
Global Int.-Term Taxables (Hedged)	50% sovereign and 50% investment-grade corporate debt of developed countries of 7-year maturity	30%
Int.-Term TIPS	US TIPS of 7-year maturity	30%
High Yield	Taxable bonds of 7-year maturity with credit characteristics of CSFB High Yield Index II	30%
Global Large-Cap (Unhedged)	MSCI World Index (NDR) Index	15%
US Diversified	S&P 500 Index	15%
US Value	S&P/Barra Value Index	15%
US Growth	S&P/Barra Growth Index	15%
US Mid-Cap	Russell Mid-Cap Index	15%
US Small-/Mid-Cap	Russell 2500 Index	15%
US Small-Cap	Russell 2000 Index	15%
Developed International	MSCI EAFE Index (Unhedged)	15%
Emerging Markets	MSCI Emerging Markets Index	20%
Global REITs	NAREIT Index	30%
Real Assets	1/3 NAREIT, 1/3 MSCI ACWI Commodity Producer Index, 1/3 DJ-UBS Commodity Futures Index	30%
Diversified Hedge Fund	Diversified Hedge Fund Asset Class	33%

Notes on Wealth Forecasting System

3. **Volatility**

   Volatility is a measure of dispersion of expected returns around the average. The greater the volatility, the more likely it is that returns in any one period will be substantially above or below the expected result. The volatility for each asset class used in this analysis is listed on the Capital-Market Projections page. In general, two-thirds of the returns will be within one standard deviation. For example, assuming that stocks are expected to return 8.0% on a compounded basis and the volatility of returns on stocks is 17.0%, in any one year it is likely that two-thirds of the projected returns will be between (8.9)% and 28.8%. With intermediate government bonds, if the expected compound return is assumed to be 5.0% and the volatility is assumed to be 6.0%, two-thirds of the outcomes will typically be between (1.1)% and 11.5%. Bernstein's forecast of volatility is based on historical data and incorporates Bernstein's judgment that the volatility of fixed-income assets is different for different time periods.

4. **Technical Assumptions**

   Bernstein's Wealth Forecasting System is based on a number of technical assumptions regarding the future behavior of financial markets. Bernstein's Capital Markets Engine is the module responsible for creating simulations of returns in the capital markets. These simulations are based on inputs that summarize the current condition of the capital markets as of December 31, 2014. Therefore, the first 12-month period of simulated returns represents the period from December 31, 2014, through December 31, 2015, and not necessarily the calendar year of 2015. A description of these technical assumptions is available on request.

5. **Expenses and Spending Plans (Withdrawals)**

   All results are generally shown after applicable taxes and after anticipated withdrawals and/or additions, unless otherwise noted. Liquidations may result in realized gains or losses, which will have capital-gains tax implications.

6. **Tax Implications**

   Before making any asset-allocation decisions, an investor should review with his/her tax advisor the tax liabilities incurred by the different investment alternatives presented herein, including any capital gains that would be incurred as a result of liquidating all or part of his/her portfolio, retirement-plan distributions, investments in municipal or taxable bonds, etc. Bernstein does not provide tax, legal, or accounting advice. In considering this material, you should discuss your individual circumstances with professionals in those areas before making any decisions.

Notes on Wealth Forecasting System

7. Tax Rates

Bernstein's Wealth Forecasting Analysis has used the following tax rates for this analysis:

Taxpayer	Scenario	Start Year	End Year	Federal Income-Tax Rate	Federal Capital Gains Tax Rate	State Income-Tax Rate	State Capital-Gains Tax Rate	Tax Method Type
Client	All	2015	2024	43.4%	23.8%	0.0%	0.0%	Top Marginal

The federal income tax rate represents Bernstein's estimate of either the top marginal tax bracket or an "average" rate calculated based upon the marginal rate schedule. The federal capital-gains tax rate is represented by the lesser of the top marginal income tax bracket or the current cap on capital gains for an individual or corporation, as applicable. Federal tax rates are blended with applicable state tax rates by including, among other things, federal deductions for state income and capital-gains taxes. The state income tax rate represents Bernstein's estimate of the "average" rate calculated based upon the applicable state's marginal tax schedule. Where an applicable state tax code permits the exclusion of a portion of capital-gains from gross income for purposes of calculating state income tax, such exclusions have been included in the calculation.